CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (CE) Examination

April 2013

CE209 Design & Analysis of Algorithms

Date: 24.04.2013, Wednesday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) What is Order of Growth? Explain with the help of an example. [02]

(b) Write the differences between Dynamic Programming and Divide and Conquer strategy. [02]

(c) Why should we usually concentrate on finding the worst-case running time only? [03]

Q - 2 (a) Prove that recursive solution of Binomial Coefficient takes time of $\Theta(2^n)$. [04]

(b) Give the usefulness of Divide and Conquer Method for sorting and searching algorithms. [04]

(c) Solve the following Recurrences using Master Method: [06]

1. $T(n) = 2T(n/2) + n^3$
2. $T(n) = 7T(n/2) + n^2$

OR

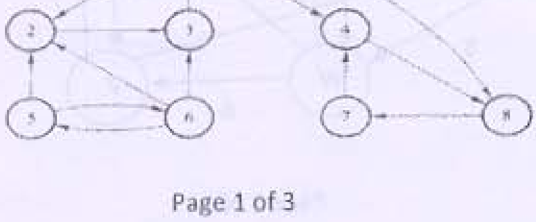
Q - 2 (a) Prove that recursive solution of Fibonacci series takes time of $O(2^n)$. [04]

(b) List out the various string matching algorithms. Compare the preprocessing time and matching time of those algorithms. Give the application of String Matching algorithms. [04]

(c) Use Recursion Tree Method to solve the recurrence $T(n) = T(n/3) + T(2n/3) + cn$ and verify your answer using substitution method. [06]

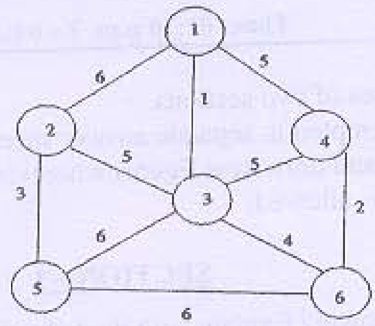
Q - 3 **Attempt Any Two** [14]

(a)



For the given graph, write the sequences of nodes according to Breadth First Search and Depth First Search. [Source node is 1]. Give the applications of BFS and DFS.

(b) Is Kruskal's algorithm greedy? Justify your answer and find the Minimum Spanning Tree by applying Kruskal's algorithm.



(c) Using Dynamic programming, find an optimum order of a matrix chain product whose sequence of dimensions is <5,10,3,5,6,2>.

SECTION – II

Q - 4 (a) What kinds of problems can be solved by Algorithms? [02]

(b) Give similarities and differences between Branch & Bound and Backtracking. [02]

(c) Compare the two functions n^2 and $2^n/4$ for various values of n. Determine when the second becomes larger than the first. [03]

Q - 5 (a) Give general characteristics of Greedy Algorithms. [04]

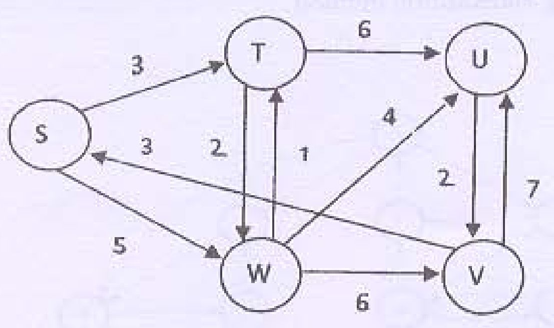
(b) Define polynomial time algorithm, P, NP and NP-Complete problems. [04]

(c) Write the algorithm for string matching using Finite-Automata and Construct the string matching automata for the pattern P= aabab. [06]

OR

Q - 5 (a) Explain optimal substructure and overlapping subproblems with respect to Dynamic Programming with example. [04]

(b) Output the sequence of vertices identified by the Dijkstra's algorithm for single source shortest path when the algorithm is started at node S for the given weighted directed graph. [05]



(c) How many spurious hits does the Rabin-Karp matcher encounter in the text T=3141592653589793 when looking for the pattern = 26? Take value of modulo q=11. [05]

Q – 6 **Attempt Any Two** [14]

(a) What is the best case and worst case complexity for insertion sort? Prove it.

(b) Solve the following Task Assignment problem using Branch & Bound Technique. Show your calculation.

	1	2	3	4	5
A	10	5	13	15	16
B	3	9	18	13	6
C	10	7	2	2	2
D	7	11	9	7	12
E	7	9	10	4	12

(c) Solve the following 0/1 Knapsack problem using Backtracking. There are four objects whose weights and values are given in the following arrays:

Weight w = {2,3,4,5} and Value v = {3,5,6,10}

Find out the optimal knapsack objects for weight capacity of 8 units.

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